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EMBEDDED THERMOELECTRIC COOLER USING THERMALLY ANISOTROPIC MESAS FOR POWER DEVICE HEAT GENERATING SOURCE TEMPERATURE REDUCTION

Abstract

A microelectronic device including a substrate having a semiconductor material containing an embedded thermoelectric cooler with thermally anisotropic mesas between the cold terminal and the hot terminal of the embedded thermoelectric cooler adjacent to a heat source; the adjacent embedded thermoelectric cooler providing a temperature reduction for the heat source resulting in increased safe operating area (SOA) for the microelectronic device. The thermally anisotropic mesas are formed in parallel with deep trenches used as isolation in the microelectronic device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Divisional of U.S. patent application Ser. No. 17/836,731, filed on Jun. 9, 2022, which claims the benefit from U.S. Provisional Patent Application No. 63/291,340 (Texas Instruments Docket No. T101457US01), filed on Dec. 17, 2021, both of which are hereby incorporated herein by reference in their entirety.

FIELD

[0002] This disclosure relates to the field of microelectronic devices. More particularly, but not exclusively, this disclosure relates to an embedded silicon based thermoelectric cooler for microelectronic device heat generating source temperature reduction.

BACKGROUND

[0003] As microelectronic devices trend towards higher power and smaller size, thermal management become more difficult. Local hot spots from heat generating sources may degrade device performance and reliability. In next generation of devices, there is a desire to improve performance, decrease die size, and provide better thermal management of local hot spots from heat generating sources through methods of cooling the microelectronic device.

SUMMARY

[0004] This summary is provided to introduce a brief selection of disclosed concepts in a simplified form that are further described below in the detailed description including the drawings provided. This summary is not intended to limit the claimed subject matter's scope.

[0005] Disclosed examples include microelectronic devices including an integrated silicon based embedded thermoelectric cooler with anisotropic thermal mesas for thermal insulation between thermopiles of a cold terminal and thermopiles of a hot terminal of the embedded thermoelectric cooler for thermal management and hot spot reduction of heat generating sources of a microelectronic device. Disclosed examples provide an associated process flow for forming such integrated silicon based microelectronic devices.

Description

BRIEF DESCRIPTION OF THE VIEWS OF THE DRAWINGS

[0006] FIG. 1A is a cross section of an example microelectronic device having an integrated thermoelectric cooler containing thermally anisotropic mesas and a heat generating source (LDMOS transistor in this example).

[0007] FIG. 1B is a detailed cross section of a thermally anisotropic mesa and laterally adjacent deep trenches with a trench liner.

[0008] FIG. 1C is a detailed cross section of a thermally anisotropic mesa and laterally adjacent deep trenches without a trench liner.

[0009] FIG. 1D is a detailed cross section of a thermally anisotropic mesa and laterally adjacent deep trenches which include an air gap.

[0010] FIG. 1E is a perspective view of thermally anisotropic mesas and laterally adjacent deep trenches as well as thermopiles and laterally adjacent trenches.

[0011] FIG. 1F is a perspective view of thermally anisotropic mesas and laterally adjacent deep trenches as well as thermopiles and laterally adjacent trenches.

[0012] FIG. 2 is a top-down view of an example microelectronic device having a power transistor and an embedded thermoelectric cooler in an example configuration.

[0013] FIG. 3 presents a flowchart of an example method of forming the microelectronic device of FIG. 1A.

[0014] FIG. 4 is a graph presenting a thermal simulation of the reduction in heat generating source temperature when an embedded thermoelectric cooler waveform is applied.

DETAILED DESCRIPTION

[0015] The present disclosure is described with reference to the attached figures. The figures are not drawn to scale and they are provided merely to illustrate the disclosure. Several aspects of the disclosure are described below with reference to example applications for illustration. It should be understood that numerous specific details, relationships, and methods are set forth to provide an understanding of the disclosure. The present disclosure is not limited by the illustrated ordering of acts or events, as some acts may occur in different orders and/or concurrently with other acts or events. Furthermore, not all illustrated acts or events are required to implement a methodology in accordance with the present disclosure.

[0016] In this disclosure and the claims that follow, unless stated otherwise and/or specified to the contrary, any one or more of the layers set forth herein can be formed in any number of suitable ways, such as with spin-on techniques, sputtering techniques (e.g., Magnetron and/or ion beam sputtering), thermal growth techniques or deposition techniques such as chemical vapor deposition (CVD), physical vapor deposition (PVD), PECVD, or atomic layer deposition (ALD), for example. As another example, silicon nitride may be a silicon-rich silicon nitride or an oxygen-rich silicon nitride. Silicon nitride may contain some oxygen, but not so much that the materials dielectric constant is substantially different from that of high purity stoichiometric silicon nitride.

[0017] It is noted that terms such as top, bottom, and under may be used in this disclosure. These terms should not be construed as limiting the position or orientation of a structure or element, but should be used to provide spatial relationship between structures or elements. For the purposes of this disclosure, the term lateral refers to a direction parallel to a plane of the instant top surface of the microelectronic device and the term vertical is understood to refer to a direction perpendicular to the plane of the instant top surface of the microelectronic device.

[0018] Thermal management and heat generating source temperature reduction of high-power elements are important for high power microelectronic device reliability and improving the safe operating area (SOA) of microelectronic devices. An embedded thermoelectric cooler may aid in thermal management by placing the cold terminal of the embedded thermoelectric cooler laterally adjacent to a heat generating area of the high-power element of the microelectronic device. The embedded thermoelectric cooler contains a cold terminal which removes heat from near the heat generating source and a hot terminal more distant from the heat generating source than the cold terminal which spreads the heat to other areas of the microelectronic device. An array of thermally anisotropic mesas is between the cold terminal and the hot terminal of the thermoelectric cooler. The thermally anisotropic mesas between the hot terminal and cold terminal of the embedded thermoelectric cooler provide an advantageous thermal barrier which allows heat to be moved between the hot terminal and cold terminal through an interconnect system while minimizing thermal leakage between the cold terminal and the hot terminal through the silicon. The embedded thermoelectric cooler reduces the highest temperatures at the heat generating source and distributes that heat across the rest of the microelectronic device. The embedded thermoelectric cooler may be powered by using a transient electrical waveform from an electronic circuit within the microelectronic device. By modulating the shape and timing of the transient electrical waveform, the heat generating source temperature reduction may be modulated and adapted to the use conditions of the microelectronic device. By placing of the embedded thermoelectric cooler with

the cold terminal adjacent to the heat generating source, a means to dissipate the abrupt temperature changes from local hot spots during transient heating events is achieved. Having at least 0.10 atomic percent germanium in the thermally anisotropic mesas reduces the thermal conduction between the upper end of the thermally anisotropic mesas at the top surface of the silicon and lower end of the thermally anisotropic mesas in the base wafer which may advantageously improve the thermal barrier between the cold terminal and the hot terminal of the embedded thermoelectric cooler. The thermally anisotropic mesas are formed concurrently with deep trenches in current process flows which include deep trench processing for device isolation.

[0019] FIG. 1A is a cross section of an example microelectronic device **100** comprising an embedded thermoelectric cooler **102** and a heat source **104**. The microelectronic device **100** includes an array of thermally anisotropic mesas **122** between the thermopiles **128** of the cold terminal **130** and the thermopiles **128** of the hot terminal **132** of the embedded thermoelectric cooler **102**. In FIG. 1A, the thermopiles **128** of the cold terminal **130** are in a region of p-type body well **136** (second conductivity type). The thermopiles of the hot terminal **132** may be in a n-type well region **138** (first conductivity type). The thermopiles **128** of the cold terminal **130** and the thermopiles **128** of the hot terminal **136** being of opposite conductivity. While FIG. 1A shows the cold terminal **130** in a shared p-type body well **136** with the heat generating source **104**, the cold terminal **130** may be in an independent well region of the second conductivity type (not specifically shown). The thermopiles **128**, and the thermally anisotropic mesas **122** are less than 60 nm wide at the surface **105** of the silicon with no dimensional restriction in length.

[0020] In FIG. 1A, the heat generating source **104** is shown as a laterally diffused MOS transistor (LDMOS) transistor. Other examples of heat generating sources (not specifically shown) include, a drain extended metal oxide semiconductor (DEMOS) transistor, a bipolar junction transistor, a junction field effect transistor, a gated bipolar, a gated unipolar semiconductor device, an insulated gate bipolar transistor (IGBT), a silicon controlled rectifier (SCR), a metal oxide semiconductor (MOS)-triggered SCR, a MOS-controlled thyristor, a gated diode, a resistor, an amplifier, and a Schottky diode. The remaining discussion refers to an LDMOS transistor as the heat generating source **104**. One of ordinary skill in the art will understand to replace the elements of the LDMOS transistor with the appropriate elements of the desired heat generating device. Although NMOS LDMOS transistors are described herein as the heat generating source **104**, it is clear that p-channel (PMOS) LDMOS transistors can be formed when n-doped regions are substituted by p-doped regions and p-doped regions are substituted by n-doped regions.

[0021] The microelectronic device **100** includes a substrate **106** that has silicon semiconductor material **108**, referred to herein as the silicon **108**. The substrate **106** may be, for example, part of a bulk semiconductor wafer, part of a semiconductor wafer with an epitaxial layer, part of a silicon-on-insulator (SOI) wafer, or other structure suitable for forming the microelectronic device **100**.

[0022] The substrate **106** may include an optional n-type buried layer (NBL) **112** on a base wafer **110**. The base wafer **110** may be p-type with a dopant concentration of 1×10^{17} atoms/cm³ to 1×10^{18} atoms/cm³, for example. Alternatively, the base wafer **110** may be lightly doped, with an average dopant concentration below 1×10^{16} . The NBL **112** may be 2 microns to 10 microns thick, by way of example, and may have a dopant concentration of 1×10^{16} atoms/cm³ to 1×10^{17} atoms/cm³. The substrate **106** may include an epitaxial layer **114** of silicon on the NBL **112**. The epitaxial layer **114** is part of the silicon **108**, and may be 2 microns to 12 microns thick, for example. The epitaxial layer **114** may be p-type, with a dopant concentration of 1×10^{15} atoms/cm³ to 1×10^{16} atoms/cm³, by way of example. In versions of this example in which the substrate **106** lacks the NBL **112**, the epitaxial layer **114** may be directly on the base wafer **110**. The substrate **106** may also include an optional blanket p-type buried layer (PBL) (not specifically shown) in the epitaxial layer **114**. The PBL layer may be 1 micron to 5 microns thick, and may have a dopant concentration of 1×10^{15} atoms/cm³ to 3×10^{17} atoms/cm³, by way of example.

[0023] The thermally anisotropic mesas **122**, deep trenches **118**, and deep trench gap fill material **120** of the thermally anisotropic mesa region **133** are between the cold terminal **130** of the embedded thermoelectric cooler **102** and hot terminal **132** of the embedded thermoelectric cooler **102**. The thermally anisotropic mesas **122** and deep trenches **118** extend from the top surface **105** of the substrate **106** into silicon **108**. The deep trenches **118** may extend through the NBL **112** into the base wafer **110**. The thermally anisotropic mesas **122** advantageously provide thermal insulation between the thermopiles **128** of the cold terminal **130** of the embedded thermoelectric cooler **102** and the thermopiles **128** of the hot terminal **132** of the embedded thermoelectric cooler **102**.

[0024] The deep trench gap fill material **120** may be filled with a deep trench sidewall dielectric layer **170** and an electrically conductive deep trench fill material **172** as shown in FIG. 1B, a deep trench fill material **174** without a deep trench sidewall dielectric layer **170** as shown in FIG. 1C, or by an air gap **178** as shown in FIG. 1D.

[0025] The microelectronic device **100** of FIG. 1A may contain a field oxide **126** in a field oxide trench **127** (configured in this example as shallow trench isolation) which provides isolation for the LDMOS transistor. The field oxide **126** also provides isolation between the thermopiles **128** of the embedded thermoelectric cooler **102**. Additional components in the silicon **108** of the embedded thermoelectric cooler **102** include a deep well region **124**, a p-type body well region **136**, and a n-type well region **138**. While the example in FIG. 1A is shown with the thermopiles **128** of the cold terminal **130** in the p-type body well region **136** and the thermopiles **128** of the hot terminal **132** in the n-type well region **138**, the case where the thermopiles **128** of the cold terminal **130** are in a n-type well region and the thermopiles **128** of the hot terminal **132** are in a p-type well region is also within the scope of the disclosure.

[0026] The heat generating source **104** example LDMOS of FIG. 1A contains an n-drift region (NDRIFT) **134**, a p-type body well region **136**, a gate dielectric layer **140**, a gate electrode **142**, sidewall spacers **144**, a source region **146**, a drain region **148**, and a back gate region **150**. The p-type body well region **136** may be a shared well region between the embedded thermoelectric cooler **102** and the heat generating source **104**.

[0027] Above the top surface **105** of the embedded thermoelectric cooler **102** and the heat generating source **104** a silicide (not specifically shown) may be formed over silicon **108** exposed at the top surface **105** and over the gate electrode **142** to improve resistance to contacts which are formed through the pre-metal dielectric (PMD) layer **160**. The PMD layer **160** is over the top surface **105**, the gate electrode, and the field oxide **126**. The source region **146**, drain region **148**, back gate region **150**, and the gate electrode **142** are electrically connected through contacts and a heat generating source interconnect system **158**. The embedded thermoelectric cooler **102** is connected to thermoelectric terminals **154** through contacts and interconnects, and the hot terminal **132** of the embedded thermoelectric cooler **102** is connected to the cold terminal **130** of the embedded thermoelectric cooler **102** through an embedded thermoelectric cooler interconnect system **156**. The embedded thermoelectric cooler interconnect system **156** contains one or more metal levels, one or more via levels, and may have a first inter-level dielectric (ILD) **161**, and may have a second ILD **163**.

[0028] FIG. 1B is a detailed section of a thermally anisotropic mesa **122** in the epitaxial layer **114** and the silicon **108** with deep trenches **118** adjacent to the thermally anisotropic mesa **122** highlighted in FIG. 1A. FIG. 1B shows an embodiment of the disclosure in which the deep trenches **118** adjacent to the thermally anisotropic mesa **122** are formed concurrently with deep trenches in other parts of the semiconductor device **100** using existing semiconductor processes for deep trench isolation and deep trench capacitors. The deep trenches **118** are configured with a deep trench sidewall dielectric layer **170** and an electrically conductive deep trench fill material **172**. The electrically conductive deep trench fill material **172** may be a conductive material such as polysilicon. In the example shown in FIG. 1B, the deep trenches **118** are etched through the NBL

112 and into the base wafer **110**. The deep trenches may be surrounded between the top surface **105** to the NBL **112** by the deep well region **124**. The PMD layer **160** is over the top surface **105** of the silicon **108**.

[0029] FIG. **1C** is a detailed section of a thermally anisotropic mesa **122** in the epitaxial layer **114** and the silicon **108** with deep trenches **118** adjacent to the thermally anisotropic mesa **122** highlighted in FIG. **1A**. FIG. **1C** shows a second embodiment of the disclosure in which the deep trenches **118** adjacent to the thermally anisotropic mesa **122** are configured with a deep trench fill material **174** directly contacting the silicon **108**. The deep trench fill material **174** is a dielectric material such as silicon dioxide or silicon nitride. In the example shown in FIG. **1C**, the deep trenches **118** are etched through the NBL **112** and into the base wafer **110**. The deep trench may be surrounded between the top surface **105** to the NBL **112** by the deep well region **124**. The PMD layer **160** is over the top surface **105** of the silicon **108**.

[0030] FIG. **1D** is a detailed section of a thermally anisotropic mesa **122** in the epitaxial layer **114** and the silicon **108** with deep trenches **118** adjacent to the thermally anisotropic mesa **122** highlighted in FIG. **1A**. FIG. **1D** shows a third embodiment of the disclosure in which the deep trenches **118** adjacent to the thermally anisotropic mesa **122** are configured with a poor gap fill dielectric **176**, and an air gap **178** within the deep trenches **118** below the poor gap fill dielectric **176**. In the example shown in FIG. **1D**, the deep trenches **118** are etched through the NBL **112** and into the base wafer **110**. The deep trench may be surrounded between the top surface **105** to the NBL **112** by the deep well region **124**. The PMD layer **160** is over the top surface **105** of the silicon **108**. The PMD layer **160** is over the top surface **105** of the silicon **108**.

[0031] FIG. **1E** shows a perspective view of the embedded thermoelectric cooler **102** of the microelectronic device **100**. For clarity, the implants, deep trench fill, field oxide, PMD layer **160**, contacts, and interconnects have been removed. The perspective view shows the thermally anisotropic mesas **122** and thermopiles **128** as linear mesas running the length of the embedded thermoelectric cooler **102**. Likewise, the deep trenches **118** and field oxide trenches **127** are linear trenches in the silicon **108**. The thermally anisotropic mesas **122** may have a top width less than 60 nm and have a height of at least 10 microns. The thermopiles **128** may have top width of less than 60 nm and a height greater than 250 nm.

[0032] FIG. **1F** shows a perspective view of a second embodiment of the embedded thermoelectric cooler **102** of the microelectronic device **100**. For clarity, the implants, electrically conductive deep trench fill material **172**, field oxide **126**, PMD layer **160**, contacts, and interconnects have been removed. The perspective view shows the thermally anisotropic mesas **122** and thermopiles **128** as pillars of mesas running the length of the embedded thermoelectric cooler **102**. Likewise, the deep trenches **118** and the field oxide trenches **127** in the silicon **108** surround each of the thermally anisotropic mesas **122** and thermopiles **128**. The thermally anisotropic mesas **122** may have a top width of less than 60 nm and a height greater than 10 microns. The thermopiles **128** may have a top width of less than 60 nm and a height of greater than 250 nm.

[0033] FIG. **2** shows a top-down view of a microelectronic device **200** containing an embedded thermoelectric cooler **202** and a heat generating source **204** in the silicon **208**. In FIG. **2**, the heat generating source **204** is an LDMOS transistor. In the example shown in FIG. **2**, The embedded thermoelectric cooler **202** is placed at four locations adjacent to the heat generating source **204**, but could be placed in other geometries adjacent to the heat generating source **204**. Additionally, placing the embedded thermoelectric cooler **202** in less than or more than four locations adjacent to the heat generating source **204** is possible. The heat generating source **204** LDMOS transistor consists of a source region **246**, a drain region **248**, a gate electrode **242**, and field oxide **226**. Adjacent to the source region **246** on each side of the heat generating source **204** is a cold terminal **230** of the embedded thermoelectric cooler **202**. Adjacent to the cold terminal **230** of the embedded thermoelectric cooler **202**, and between the cold terminal **230** and the hot terminal **232** is a region containing the thermally anisotropic mesas **233**. The thermally anisotropic mesas **233** may be

continuous around the heat generating source **204** as shown in FIG. 2, or they may be segmented (not specifically shown). The hot terminal **232** of the embedded thermoelectric cooler **202** is within a region of n-well **238** and the cold terminal **230** is within a region of p-well **236**. Additionally, the microelectronic device **200** contains an electronic circuit **280**, also in the silicon **208** which generates a current with a waveform to power the embedded thermoelectric cooler **202**.

[0034] FIG. 3 refers to flowchart of an example method **300** for forming the heat generating source **104** and embedded thermoelectric cooler **102** of FIG. 1A. Structural elements referred to in the steps of the method **300** are shown in FIG. 1A.

[0035] The method **300** includes step **302**, which may include optionally forming the NBL **112** on the base wafer **110**. The NBL **112** may be formed by forming a hard mask, not explicitly shown, over the base wafer **110** that exposes the base wafer **110** in an area for the NBL **112**. N-type dopants, such as antimony and optionally arsenic are implanted into the base wafer **110** at a dose of 5×10^{14} ions/cm² to 3×10^{15} ions/cm², where exposed by the hard mask. The substrate **106** is heated to diffuse and activate the implanted n-type dopants to form the NBL **112**. After the formation of the NBL **112**, the epitaxial layer **114** may be formed by an epitaxial process. In versions of this example in which the substrate **106** lacks the NBL **112**, the epitaxial layer **114** may be formed on the base wafer **110**.

[0036] The method **300** continues with step **304** includes formation of the thermally anisotropic mesas **122**. An optional deep well region **124** may be formed in the silicon **108**, extending from the top surface **105** of the silicon **108** to the NBL **112**. The deep well region **124** is n-type in this example, but may be p-type. The deep well region **124** may be formed by implanting dopants such as phosphorus, into the silicon **108**, followed by a thermal drive to diffuse the implanted dopants to the NBL **112** and activate the implanted dopants. The deep well region **124** may have an average concentration of the dopants that is 2 to 10 times greater than an average concentration of dopants in the silicon **108** outside of the deep well region **124**.

[0037] A pad oxide layer (not specifically shown) may be formed on the top surface **105** of the silicon **108**. The pad oxide layer may include primarily silicon dioxide, may be formed by a thermal oxidation process or a chemical vapor deposition (CVD) process, and may have a thickness of 50 nanometers to 200 nanometers, by way of example. A nitride cap layer (not specifically shown) may be formed on the pad oxide layer. The nitride cap layer may include primarily silicon nitride, may be formed by a low-pressure chemical vapor deposition (LPCVD) furnace process, and may have a thickness of 100 nanometers to 500 nanometers, for example. A hard mask layer (not specifically shown) may be formed on the nitride cap layer. The hard mask layer may include primarily silicon dioxide, may be formed by a plasma enhanced chemical vapor deposition (PECVD) process, and may have a thickness of 1 micron to 3 microns. The pad oxide layer may provide stress relief between the silicon **108** and a combination of the nitride cap layer and the hard mask layer. The nitride cap layer may provide a stop layer for subsequent etch and planarization processes. The hard mask layer may provide a hard mask during a subsequent deep trench etch process to form the deep trenches **118**. Next, a trench mask (not specifically shown) may be formed on the hard mask layer with openings which expose the hard mask layer in an area for the deep trenches **118**. The trench mask may include photoresist, and may optionally include anti-reflection material such as a bottom anti-reflection coat (BARC). The trench mask may be formed by a photolithographic process.

[0038] A deep trench etch process (not specifically shown) is performed to form the deep trenches **118** and the thermally anisotropic mesas **122** in the silicon **108**. The deep trench etch process may include multiple steps. In one implementation for example, a hard mask etch may be first performed to remove the hard mask layer where exposed by the trench mask, and a silicon etch may then be performed to remove the nitride cap layer, the pad oxide layer, and the silicon **108** in regions that are exposed by the hard mask layer to form the deep trenches **118** and the thermally anisotropic mesas **122**. The deep trenches **118** and the thermally anisotropic mesas **122** extends

from the top surface **105** of the substrate **106** into the NBL **112** in this example. Additionally, a germanium implant having at least 0.10 percent germanium may be added in the silicon which encompasses the thermally anisotropic mesas. Having at least 0.10 atomic percent germanium in the thermally anisotropic mesas **122** reduces the thermal conduction between the upper end of the thermally anisotropic mesas **122** at the top surface **105** of the silicon and lower end of the thermally anisotropic mesas **122** in the base wafer **110** which may advantageously improve the thermal barrier performance of the thermally anisotropic mesas **122**.

[0039] A deep trench sidewall dielectric layer **170** shown in FIG. **1B** may be formed in the deep trenches **118**, contacting the silicon **108** of the substrate **106**. The deep trench sidewall dielectric layer **170** may include a single layer of a silicon-nitrogen compound or a silicon dioxide compound or may include multiple layers of silicon-nitrogen compounds, silicon dioxide compounds, or other dielectric materials.

[0040] An electrically conductive deep trench fill material **172** shown in FIG. **1B** may be formed in the deep trenches **118** on the deep trench sidewall dielectric layer **170**. The electrically conductive deep trench fill material **172** includes primarily silicon, and may be implemented as polycrystalline silicon, commonly referred to as polysilicon. Alternatively, the electrically conductive deep trench fill material **172** may be implemented as amorphous silicon, or semi-amorphous silicon. The electrically conductive deep trench fill material **172** may have an average concentration of dopants of 5×10^{18} cm.⁻³ and 1×10^{20} cm.⁻³.

[0041] The electrically conductive deep trench fill material **172** and the deep trench sidewall dielectric layer **170** above the top surface **105** may be removed by a planarization process, such as a chemical mechanical polish (CMP) process or an etch back process (not specifically shown). The process of removing the electrically conductive deep trench fill material **172** and the deep trench sidewall dielectric layer **170** on the top surface **105** of the substrate **106** leaves the electrically conductive deep trench fill material **172** and the deep trench sidewall dielectric layer **170** in the deep trenches **118** with thermally anisotropic mesas **122** between the deep trenches **118**. The process of removing the electrically conductive deep trench fill material **172** and the deep trench sidewall dielectric layer **170** may leave the nitride cap layer and the pad oxide layer on the top surface **105** of the substrate **106** which may subsequently be removed by a plasma or wet etch process.

[0042] The method **300** continues with step **306** includes forming the thermopiles **128** of the embedded thermoelectric cooler **102** and field oxide **126** for the heat generating source **104**. A pad oxide and a CMP stop layer (not specifically shown) are deposited over the top surface **105** of the silicon. A photomask (not specifically shown) serves the function of masking the CMP stop layer and defines regions for the thermopiles **128** and areas of field oxide **126**. The photomask may include a light sensitive organic material that is coated, exposed and developed. The photomask step is followed by a plasma etch process (not specifically shown) which removes the CMP stop layer, the pad oxide, and the epitaxial layer **114** to form the thermopiles **128** and field oxide trenches **127** for the field oxide **126**. The field oxide trench may be between 250 nm and 1000 nm in depth. The field oxide isolates the thermopiles **128** from each other as well as forms the field oxide **126** of the heat generating source **104**. Additionally, a germanium implant having at least 0.10 percent germanium may be added in the silicon which encompasses the n-type thermopiles and the p-type thermopiles.

[0043] The field oxide **126** is formed in the field oxide trenches **127** and over the CMP stop layer. The field oxide **126** may include primarily silicon dioxide, or silicon dioxide-based dielectric material, formed by one or more CVD processes alternated with etch-back processes to provide complete filling of the field oxide trench. The field oxide **126** is planarized so that the field oxide **126** does not extend over the top surface **105** of the silicon **108**. The field oxide **126** may be planarized by a CMP process (not specifically shown). After the field oxide **126** is planarized, the CMP stop layer is removed. The CMP stop layer may be removed by a wet etch process using an

aqueous solution of phosphoric acid at 140° C. to 170° C. The pad oxide may optionally be removed by a wet etch process using an aqueous solution of buffered hydrofluoric acid. The field oxide **126** forms a field oxide stress relief region for the heat generating source **104**, and isolation between the thermopiles **128** of the embedded thermoelectric cooler **102**.

[0044] When the heat generating source is an LDMOS, the method **300** may continue with step **308**. Step **308** includes forming the NDRIPT drain drift region **134** of the heat generating source **104**. A photomask (not specifically shown) is deposited and patterned with an opening where a drift region (NDRIPT) implant is to be implanted in the silicon **108** to form an NDRIPT drain drift region **134** within the exposed areas of the epitaxial layer **114**. The implant to define the NDRIPT drain drift region **134** may occur in multiple steps. In a series of implants, phosphorus may be implanted at a total dose of between $3 \times 10^{12} \text{ cm}^{-2}$ and $6.6 \times 10^{12} \text{ cm}^{-2}$ with energies between 0.5 mega-electron volts (MeV) and 2.8 MeV. Arsenic may also be implanted with a dose of between $1.5 \times 10^{12} \text{ cm}^{-2}$ and 3.0×10^{12} with an energy of between 180 kilo-electron volts (keV) and 460 keV. All of the implant process uses the same photomask to complete the formation of the NDRIPT drain drift region **134**.

[0045] The method **300** may then continue with step **310** which includes forming the p-type body well region **136** and a n-type well region **138**. A p-type body well implant process (not specifically shown) implants p-type dopants into a portion of the epitaxial layer **114** laterally adjacent to the NDRIPT drain drift region **134** to form the p-type body well region **136**. The p-type dopants may include boron. Besides boron, the p-type dopants can include indium. Indium, being a relatively large atom, has the advantage of a low diffusion coefficient relative to boron. In the case of a boron implant, the well ion implant process energy and dose used should generally be sufficient to enable formation of a channel laterally and to be suppress body NPN effects during operation of the heat generating source **104**. For example, a series of boron implants with an energy between 80 keV and 3 MeV, and doses between $4.0 \times 10^{12} \text{ cm}^{-2}$ to $1.5 \times 10^{14} \text{ cm}^{-2}$, with a tilt angle of less than 10 degrees may be used to implant the p-type body well region **136**. A pattern and implant step (not specifically shown) are used to define the n-type well region **138**. An arsenic with a dose between $6.0 \times 10^{13} \text{ cm}^{-2}$ and $8.0 \times 10^{15} \text{ cm}^{-2}$ with an energy between 60 keV to 120 keV with a tilt angle of between 0 degrees and 45 degrees may be used for the n-type well region **138** dopant.

[0046] The method **300** may then continue with step **312** which includes forming the gate dielectric layer **140** and gate electrode **142**. A gate dielectric layer **140** is first formed in a high temperature furnace operation or a rapid thermal process (not specifically shown). The gate dielectric layer **140** thickness can range from approximately 3 nm to 15 nm for silicon dioxide or a silicon oxynitride (SiON) gate dielectric that is slightly thinner but with a higher dielectric constant than that of silicon dioxide, which is about 3.9, by way of example. After the gate dielectric layer **140** is formed, a layer used to form the gate electrode **142** is deposited by a gate deposition process on the wafer using any of a number of silane-based precursors. Polycrystalline silicon is one example of a material for the gate electrode **142**, however a metal gate or CMOS-based replacement gate electrode process can also be used to provide the gate electrode **142**. After the gate dielectric layer **140** and the layer used for the gate electrode **142** are formed, a photomask is deposited and patterned. A plasma etch process is used to define the gate electrode **142**. After the plasma etch process is complete, photomask is removed and a wet or dry process is used to clean the wafer surface.

[0047] The method **300** may then continue with step **314** includes forming sidewall spacers **144** on the gate electrode **142**. The sidewall spacers **144** may be formed by forming one or more conformal layers of dielectric material over the top surface **105** of the silicon **108** including the gate electrode **142**. The dielectric material is subsequently removed from horizontal surfaces, that is, surfaces generally parallel to the top surface **105** of the silicon **108**, by an anisotropic etch process such as an RIE process, leaving the dielectric material on the lateral surfaces of the gate electrode **142**. The

sidewall spacers **144** may include a dielectric material such as silicon dioxide and silicon nitride. The sidewall spacers **144** may extend 100 nanometers to 500 nanometers from the lateral surfaces of the gate electrode **142**.

[0048] The method **300** may then continue with step **316** which includes forming a source region **146**, drain region **148**, and a back gate region **150**. A series of patterning steps (not specifically shown) and an ion implantation steps (not specifically shown) are used to implant a source region **146** in the p-type body well region **136**, implant a drain region **148** in the NDRIFT drain drift region **134**, and implant a back gate region **150**. The source region **146** and drain region **148** implantation may occur in one or more steps with implant species including one or more of phosphorus and arsenic with an overall dose of between $5 \times 10^{13} \text{ cm}^{-2}$ and $4.5 \times 10^{15} \text{ cm}^{-2}$ and an energy between 2 keV and 80 keV. The ion implantation step uses an edge of the sidewall spacers **144** abutting the source region **146**, and the drain region **148** to self-align the implants for the source region **146** and the drain region **148** respectively.

[0049] The method **300** may then continue with step **318**, which includes forming the metal silicide (not specifically shown) on the top surface **105** of the silicon **108** and on the polysilicon of the gate electrode **142**. The layer of metal which forms the silicide may include platinum, tungsten, titanium, cobalt, nickel, chromium, or molybdenum, by way of example. A cap layer of titanium nitride or tantalum nitride may be formed over the layer of metal. Subsequently, the microelectronic device **100** is heated to react the layer of metal with the silicon **108** and the polysilicon of the gate electrode **142** to form the metal silicide. Unreacted metal is removed from the microelectronic device **100**, leaving the metal silicide in place. The unreacted metal may be removed by a wet etch process using an aqueous mixture of sulfuric acid and hydrogen peroxide, or an aqueous mixture of nitric acid and hydrochloric acid, by way of example. The metal silicide may provide electrical connections to the source region **146**, the drain region **148**, the back gate region **150**, the gate electrode **142**, the thermopiles **128**, and to the thermoelectric taps **152** with lower resistances compared to a similar microelectronic device **100** without a metal silicide.

[0050] The method **300** continues with step **320**, which includes forming the PMD layer **160** over the microelectronic device **100**. The PMD layer **160** may include a PMD liner (not specifically shown) over the top surface **105** of the silicon **108**. The PMD liner may be formed in one or more sublayers by one or more CVD processes. The PMD liner is non-conducting. A silicon nitride PMD liner sublayer may be formed by a low pressure chemical vapor deposition (LPCVD) process using dichlorosilane and ammonia, or by a PECVD process using bis(tertiary-butyl-amino) silane (BTBAS). A silicon dioxide PMD liner sublayer may be formed by a PECVD process using tetraethyl orthosilicate (TEOS), formally named tetraethoxysilane. A silicon oxynitride PMD liner sublayer may be formed by a PECVD process using BTBAS and TEOS.

[0051] The PMD layer **160** may be formed over the PMD liner if present by one or more dielectric deposition processes, including a PECVD process using TEOS, a high-density plasma (HDP) process, or a high aspect ratio process (HARP) using TEOS and ozone, by way of example. The PMD layer **160** may be planarized by an oxide CMP process.

[0052] The method **300** continues with step **322**, which includes forming the contacts through the PMD layer **160**. The contacts create electrical connections to the embedded thermoelectric cooler **102** and the heat generating source **104**. The contact holes may be formed by etching contact holes through the PMD layer **160** and the PMD liner if present to expose the metal silicide. In one version of step **322**, the contacts may be formed by sputtering titanium to form a titanium adhesion layer, followed by forming the titanium nitride diffusion barrier using reactive sputtering, a CVD process or an ALD process. The tungsten core may be formed by a MOCVD process using tungsten hexafluoride (WF_6) reduced by silane initially and hydrogen after a layer of tungsten is formed on the titanium nitride diffusion barrier. The tungsten, titanium nitride, and titanium are subsequently removed from a top surface of the PMD layer **160** by an etch process, a tungsten CMP process, or a combination of both, leaving the contacts extending to the top surface of the

PMD layer **160**. In another version of step **322**, the contacts may be formed by a selective tungsten deposition process which fills the contact holes with tungsten from the bottom up, forming the contacts with a uniform composition of tungsten. Other methods of forming the contacts are within the scope of this step **322**.

[0053] The method **300** continues with step **324** which includes formation of interconnects. After the formation of contacts, the interconnects are formed over the contacts. In versions of this example in which the interconnects have an etched aluminum structure, the interconnects may be formed by depositing the adhesion layer, an aluminum layer, and an anti-reflection layer, forming an etch mask, (not explicitly shown) followed by an RIE process to etch the anti-reflection layer, the aluminum layer, and the adhesion layer where exposed by the etch mask, and subsequently removing the etch mask. The first level of interconnects form a heat generating source interconnect system **158** and the thermoelectric taps **152**. For the embedded thermoelectric cooler interconnect system **156**, of this example, three levels of interconnects are used. After the first level of metal is deposited, the first ILD **161** is deposited. After the formation of the first ILD **161**, a first level of vias is formed in a process similar to the formation of the contacts. After the formation of the first via level, a second metal level is formed by a similar process to the formation of the first metal level. A second ILD **163**, a second level of vias and a third level of metallization are formed by processes similar to the formation of the first level of vias and the second level of metal.

[0054] In versions of this example in which the interconnects have a damascene structure, the interconnects may be formed by forming an intra-metal dielectric (IMD) layer (not specifically shown) on the PMD layer **160**, and etching the interconnect trenches through the IMD layer to expose the contacts. A barrier liner (not specifically shown) may be formed by sputtering tantalum onto the IMD layer and the PMD layer **160** which is exposed and contacts, and forming tantalum nitride on the sputtered tantalum by an ALD process. A copper fill metal may be formed by sputtering a seed layer, not explicitly shown, of copper on the barrier liner, and electroplating copper on the seed layer to fill the interconnect trenches. Copper and barrier liner metal are subsequently removed from a top surface of the IMD layer **181** by a copper CMP process. Additional layers of interconnects are formed by repetitive steps of ILD formation, via formation and interconnect formation

[0055] In versions of this example in which the interconnects have a plated structure, the interconnects may be formed by sputtering the adhesion layer, containing titanium, on the PMD layer **160** and contacts, followed by sputtering a seed layer, not explicitly shown, of copper on the adhesion layer. A plating mask is formed on the seed layer that exposes areas for the interconnects. The interconnects are formed by electroplating copper on the seed layer where exposed by the plating mask. The plating mask is removed, and the seed layer and the adhesion layer are removed by wet etching between the interconnects. Additional layers of interconnects are formed by repetitive steps of ILD formation, via formation and interconnect formation.

[0056] FIG. **4** shows a graph showing a reduction in maximum temperature of the heat generating source **104** shown in FIG. **1A** when the embedded thermoelectric cooler **102** is in the on state when the heat generating source **104** is in an on state versus when the embedded thermoelectric cooler **102** is not in the on state when the heat generating source **104** is in an on state. The addition of an embedded thermoelectric cooler **102** and application of a current using a waveform to the embedded thermoelectric cooler **102** to cool the heat generating source **104** is advantageous to improve the SOA of the microelectronic device **100**. The uppermost trace in the graph is a simulated waveform of current supplied to the heat generating source **104**, the second upper trace is a waveform of the current supplied by an electronic circuit **280** to the embedded thermoelectric cooler **102** of FIG. **1A**. While the second upper trace is a triangular waveform, the waveform could be a square waveform, a rectangular waveform, a sinusoidal waveform, or a sawtooth waveform. Other waveforms are within the scope of the disclosure. The third trace (squares) is a simulation of the heat generated at the heat generating source **104** without an applied current to the embedded

thermoelectric cooler **102**, and the fourth trace (circles) is a simulation of the heat at the heat generating source **104** when a current is applied to the embedded thermoelectric cooler **102**. The graph shows that the peak temperature of the heat generating source **104** is lower when a current waveform is applied to the embedded thermoelectric cooler **102** compared to the peak temperature of the heat generating source **104** when a current waveform is not applied to the embedded thermoelectric cooler **102**. The current waveform of the heat generating source **104** shown in the uppermost trace is a square waveform. The current waveform of the embedded thermoelectric cooler **102** is offset in time from the application of current to the heat generating source **104**. Variation of the length of time the embedded thermoelectric cooler waveform is applied and the offset in time between the application of current to embedded thermoelectric cooler **102** and the heat generating source **104** are within the scope of this disclosure.

[0057] While various examples of the present disclosure have been described above, it should be understood that they have been presented by way of example only and not limitation. Numerous changes to the disclosed examples can be made in accordance with the disclosure herein without departing from the spirit or scope of the disclosure. Thus, the breadth and scope of the present invention should not be limited by any of the above-described examples. Rather, the scope of the disclosure should be defined in accordance with the following claims and their equivalents.

Claims

1. A method, comprising: forming a cold terminal and a hot terminal of a thermoelectric cooler in a substrate, wherein the cold terminal includes first conductivity type thermopiles and the hot terminal includes second conductivity type thermopiles; forming an array of thermally anisotropic mesas between the cold terminal and the hot terminal; forming a gap fill region between the thermally anisotropic mesas of the array of thermally anisotropic mesas; forming a heat generating source laterally adjacent to the cold terminal of the thermoelectric cooler; forming a pre-metal dielectric (PMD) layer over the substrate; forming contacts in the PMD layer to the cold terminal and to the hot terminal of the thermoelectric cooler, and to the heat generating source; and forming a plurality of interconnects connecting the contacts to the thermoelectric cooler and to the heat generating source.
2. The method of claim 1, wherein the heat generating source is a laterally diffused metal oxide semiconductor (LDMOS) transistor.
3. The method of claim 1, wherein the array of thermally anisotropic mesas between the first conductivity type thermopiles and the second conductivity type thermopiles corresponds to a linear array.
4. The method of claim 1, wherein the array of thermally anisotropic mesas between the first conductivity type thermopiles and the second conductivity type thermopiles corresponds to an array of rectangular pillars.
5. The method of claim 1, wherein the gap fill region between the thermally anisotropic mesas of the array of thermally anisotropic mesas includes silicon dioxide.
6. The method of claim 1, wherein the gap fill region between the thermally anisotropic mesas of the array of thermally anisotropic mesas includes polysilicon.
7. The method of claim 1, wherein the gap fill region between the thermally anisotropic mesas of the array of thermally anisotropic mesas includes an air gap.
8. The method of claim 1, wherein the thermally anisotropic mesas of the array of thermally anisotropic mesas are formed in a well region.
9. The method of claim 1, wherein the hot terminal of the thermoelectric cooler and the cold terminal of the thermoelectric cooler are formed in well regions of opposite conductivity.
10. The method of claim 1, wherein the cold terminal of the thermoelectric cooler is in a well region shared with a body well of the heat generating source.

11. The method of claim 1, further comprising: implanting germanium into the array of thermally anisotropic mesas with a dose providing at least 0.10 atomic percent germanium in the thermally anisotropic mesas of the array of thermally anisotropic mesas.
